

Numerical Algorithms Applied to Computational Quantum Chemistry
Grading Rubric for Homework 4

4.1 Extended Huckel Program

- i. **Correctly build the CNDO/2 Fock matrix for some simple systems (6 points)**
Read the specified format of input for some molecules composed of H, C, N, O, and F and corresponding basis sets. Correctly build and print the CNDO/2 Fock matrix with the initial guess of density matrix ($\mathbf{P}^\alpha = \mathbf{P}^\beta = \mathbf{0}$).
- ii. **Solve and print the Eigenvectors and Eigenvalues Self-Consistently (4 points):**
Solve the eigenvalue problem self-consistently. Please print the final eigenvectors and eigenvalues after convergence.
- iii. **Compute the total energy (2 points):** Calculate the total energy of the molecule based on the results you obtained from a converged SCF algorithm. Pay attention to units and unit conversions.
- iv. **Bonus Points (1 points):** Experiment with your CNDO/2 code by applying it to molecules not included in the sample inputs, such as N_2 and O_2 . Explore the chemical behaviors and properties these simulations reveal. For reference structures of common molecules, you can go to CCCBDB <https://cccbdb.nist.gov/expgeom1x.asp>. Discuss your findings and any interesting insights you gain from this exploration.